Detecting Political Manipulation as a Defensive Practice: A Transparent Baseline for Propaganda Detection

Working Paper I in Computational Political Anthropology — Version 2 (corrected)

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Code & data: github.com/politomorphism/politomorphism-defense
Revised November 2026 — supersedes Version 1

Correction note (what changed in Version 2)

> Two issues in Version 1 are corrected here, following independent reproduction audits. First, the official PropXplain English split has severe article-level leakage: the corpus contains only 250 news articles whose paragraphs are scattered across train, validation and test, so every test article also appears in training. Version 1 evaluated on that split and called the test set "held-out," which was not article-independent. Version 2 re-evaluates on an article-disjoint re-split (no article spans two sets). The baseline's headline performance is essentially unchanged (propagandistic-class F1 0.525 vs. the previously reported 0.506): performance did not collapse once the leakage was removed. Because the two test sets differ in composition, this shows the result is robust to the protocol change, not that the original split carried no overestimation at all. Second, Version 1's error analysis claimed the lexical model "systematically misses" structural rather than overt manipulation and was "in principle blind" to it. An independent blinded audit of the errors did not support this: overt lexical cues were present in most sampled false negatives, structural forms were not enriched among false negatives, and the illustrative "9/11" example in fact contains overt loaded vocabulary. That claim is therefore retracted (Section 6).

Abstract

This paper frames the automated detection of political manipulation as a defensive practice — detection, measurement, and explanation rather than generation — and establishes a transparent, reproducible baseline for it. Using the English PropXplain corpus, I train a TF-IDF and logistic-regression classifier to separate propagandistic from non-propagandistic news paragraphs. Because the corpus's official split scatters each article's paragraphs across train, validation and test (every test article also occurs in training), I re-evaluate on an article-disjoint re-split. On the article-disjoint test set (862 paragraphs; 30.6% propagandistic), the baseline attains a propagandistic-class F1 of 0.525 (precision 0.464, recall 0.606; average precision 0.527), comparable to the leaky-split figure and consistent with a weak but non-trivial detector. A matched-precision analysis shows recall falls steeply when precision is held high (recall 0.34 at precision 0.50 on the validation-selected threshold), locating the difficulty honestly. I report precision, recall and F1 on the minority (manipulation) class rather than accuracy, which is misleading under imbalance. I do not claim, as an earlier version did, that the model fails specifically on structurally rather than overtly manipulative material; an independent blinded audit did not support that claim, and it is retracted. The baseline is positioned as the defensive audit layer of the Politomorphism Engine, with the recovery of fine manipulation left explicitly open.

Keywords: propaganda detection; data leakage; article-disjoint evaluation; class imbalance; computational political anthropology; defensive auditing; Politomorphism

1. Introduction

Political manipulation is, before it is anything computational, an anthropological phenomenon: a practice by which a speaker reorganizes an audience's perceptual and moral field so that a contingent interpretation comes to feel like a natural fact. What is new is that the practice now operates at machine scale. This paper asks what a computational defense looks like, and argues it must be built around detection, measurement, and explanation rather than around a generator of manipulation wearing a defensive label — a model optimized to produce persuasive manipulation is the same artifact whatever its stated intent. The contribution here is a deliberately modest, honestly reported baseline that separates propagandistic from non-propagandistic news text, offered as a floor and as a proposed audit layer for the Politomorphism Engine — with the caveat, made explicit throughout, that a single paragraph-level classifier does not by itself validate that larger framework.

2. Corpus and the leakage problem

Experiments use the English configuration of PropXplain (Qatar Computing Research Institute), a multilingual, explanation-annotated corpus. Each paragraph carries a binary label and a natural-language explanation; this baseline uses only the text and the label. A reproduction audit established that the corpus's official train/validation/test split is not article-independent: the English partitions together contain only 250 distinct news articles, 241 of which have paragraphs in more than one split, and every one of the 225 article identifiers present in the official test set also appears in training. Ten test paragraphs are moreover exact-text duplicates of training paragraphs. Evaluating on that split does not measure generalization to unseen articles. I therefore pool all rows, remove exact-text duplicates, and re-split by article so that no article spans two sets.

3. Method

The classifier is unchanged from Version 1: TF-IDF over lowercased unigrams and bigrams (min_df = 2, sublinear term frequency), followed by logistic regression with balanced class weights, which prevents the optimizer from being rewarded for always predicting the majority class. The only change is the evaluation protocol. Pooling and de-duplicating the corpus yields 5,929 unique-text paragraphs across 250 articles; a group split by article (fixed seed) produces an article-disjoint training set of 4,366 paragraphs (1,124 propagandistic), a validation set of 701 (158 propagandistic), and a test set of 862 (264 propagandistic). The model is fit on the training set only and evaluated once on the test set; the validation set is used solely to select operating-point thresholds for the matched-precision analysis. Metrics are precision, recall and F1 on the propagandistic (positive) class, with average precision as a threshold-independent summary.

4. Results

Table 1 reports the article-disjoint results. The baseline attains a propagandistic-class F1 of 0.525 (95% article-clustered bootstrap CI [0.465, 0.581]; precision 0.464 [0.396, 0.535], recall 0.606 [0.536, 0.680]), with average precision 0.527 and accuracy 0.665. These are close to the leaky-split figures reported in Version 1 (F1 0.506), so observed performance remained comparable after the protocol change; because the two test sets differ, this indicates robustness to the change rather than proof that the original split carried no overestimation. The intervals (95% percentile intervals over 2,000 bootstrap resamples drawn at the article level) are wide: this is one weak detector on a modest test set, and nothing here should be read as a precise estimate.

Class	Precision	Recall	F1	Support
non-propagandistic	0.799	0.691	0.741	598

Class	Precision	Recall	F1	Support
propagandistic	0.464 [.40–.54]	0.606 [.54–.68]	0.525 [.47–.58]	264
Accuracy			0.665	862
Average precision (prop.)			0.527	—

Table 1. Per-class performance on the article-disjoint test partition (n = 862). Brackets on the propagandistic row are 95% article-clustered bootstrap intervals (2,000 resamples).

To show that the reported figure is not an artifact of one feature choice, Table 2 compares several lexical baselines on the same clean split. They cluster between F1 0.53 and 0.56; the reported uni+bigram model is representative rather than the strongest (character n-grams do marginally better), and all of them sit far above the trivial always-negative classifier, which detects nothing.

Lexical model (propagandistic class)	Precision	Recall	F1
Trivial (always non-propagandistic)	—	0.000	0.000
TF-IDF unigram + LR	0.469	0.602	0.527
TF-IDF char 3–5-gram + LR	0.506	0.614	0.555
TF-IDF uni+bigram + LR (reported)	0.464	0.606	0.525

Table 2. Alternative lexical baselines on the article-disjoint test set, propagandistic class. The reported model is shaded.

Accuracy (0.665) is reported but is not the headline: under this class balance a trivial always-negative classifier would score about 0.69 while detecting no manipulation, so positive-class precision and recall are the meaningful quantities. A matched-precision analysis exposes the baseline's real weakness. Decision thresholds were selected on the validation set to meet a precision target and then applied unchanged to the test set, which was not used to select them. At a validation precision target of 0.50, the achieved test precision was 0.56 and recall 0.34; at a target of 0.60, test precision 0.66 and recall 0.19; at 0.70, test precision 0.69 and recall 0.16. A validation precision target does not guarantee the same precision on test — the small gaps above show this — so these points are reported as achieved test precision with the corresponding recall, not as exact operating points. The detector can flag manipulation at modest precision but cannot sustain both high precision and useful recall, which is the honest shape of the problem.

5. Limitations

The baseline is lexical and the result is a single fixed-seed run; article-clustered bootstrap intervals for the headline metrics are reported with the code and should be consulted before reading any difference as real. The corpus is drawn from one journalistic source under a particular guideline set, so both the class distribution and the operative definition of "propagandistic" are specific to it. The upstream corpus also contains a few exact-duplicate texts carrying conflicting labels; exact-text de-duplication — performed on the pooled data before the article split, keeping the first occurrence — removes these from the clean split, so the annotation noise they reveal is noted here but does not affect the reported test set. The detector measures textual signatures only and makes no claim about intent, coordination, or effect.

Two scoping limits deserve emphasis, because they bound what the Politomorphism Engine can claim on this basis. First, this is a classification result, not a validated measurement of a construct: a score that separates corpus-labelled propaganda from non-propaganda is not yet shown to measure "manipulation density" as such, and it may partly track polarization, legitimate political criticism, or

emotional tone rather than manipulation. A manipulation map built from these predictions would, without separate construct-validity evidence (convergent and discriminant), show the distribution of the model's predictions rather than the distribution of manipulation itself. Establishing that the indicator measures the intended construct, stably and comparably across contexts, is future work, not a result claimed here. Second, and relatedly, a detector trained on human "propagandistic" labels must not be treated as an arbiter of which political positions are legitimate. At most it flags verifiable discursive techniques; political disagreement must not be recoded as manipulation, and any deployment as an audit instrument has to be designed around that distinction.

6. Retraction of the structural-failure claim

Version 1 asserted that the lexical model "systematically misses manipulation carried by presupposition, insinuation, and rhetorical equivalence" and was "in principle blind" to it, illustrating this with a passage about a "9/11" equivalence described as lacking overtly loaded vocabulary. An independent blinded audit of the model's errors does not support this claim. Overt lexical cues were present in most sampled false negatives (so the model is not failing only on non-lexical material); structural forms were not more frequent among false negatives than among true positives; no determinate structural-only false negative was found; and the "9/11" passage in fact contains overt loaded terms ("lies," "settler regime"), so it does not illustrate a purely structural miss. The audit is provisional, single-coder interpretive work, not human-validated annotation, and it does not establish the opposite claim either. But it removes the evidentiary basis for the Version 1 assertion, which is therefore withdrawn. Whether fine, structural manipulation is disproportionately missed remains an open, testable question requiring human annotation and article-held-out evaluation; it is not resolved here. This paper withdraws the unsupported claim without substituting a new one: a positive, independently annotated characterization of the errors that remain under the clean split is deferred to future work rather than asserted now.

7. Conclusion

A defensive computational practice for political manipulation should be built around detection, measurement, and explanation. This paper establishes a transparent baseline, evaluates it honestly on an article-disjoint split after correcting a leakage problem in the official split, and withdraws an earlier over-stated claim about structural failure that an audit did not support. The baseline is a floor, and the recovery of fine manipulation — sustaining recall at useful precision — is left explicitly open. This result is a binary paragraph-level classifier; it does not validate the Politomorphism framework, whose empirical validation (construct validity, independent annotation, generalization across sources and languages) remains a separate and open task that this paper only sets up.

Reproducibility

The article-disjoint split is built from the public PropXplain English partitions by pooling all rows, de-duplicating exact texts, and applying scikit-learn GroupShuffleSplit grouped by article identifier (test 15% of articles, then validation 12% of the remainder) with random seed 42. The classifier is TfidfVectorizer(lowercase, ngram_range=(1,2), min_df=2, sublinear_tf=True) followed by LogisticRegression(max_iter=2000, class_weight='balanced') under scikit-learn 1.9.1; matched-precision thresholds are selected on validation only. Per-paragraph scores and the article-clustered bootstrap (2,000 resamples) are saved with the code. Code, pinned dependencies, and this paper are maintained at github.com/politomorphism/politomorphism-defense. This version supersedes Version 1; the change log is the Correction note above.

References

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